Section 2: Number Systems Extension

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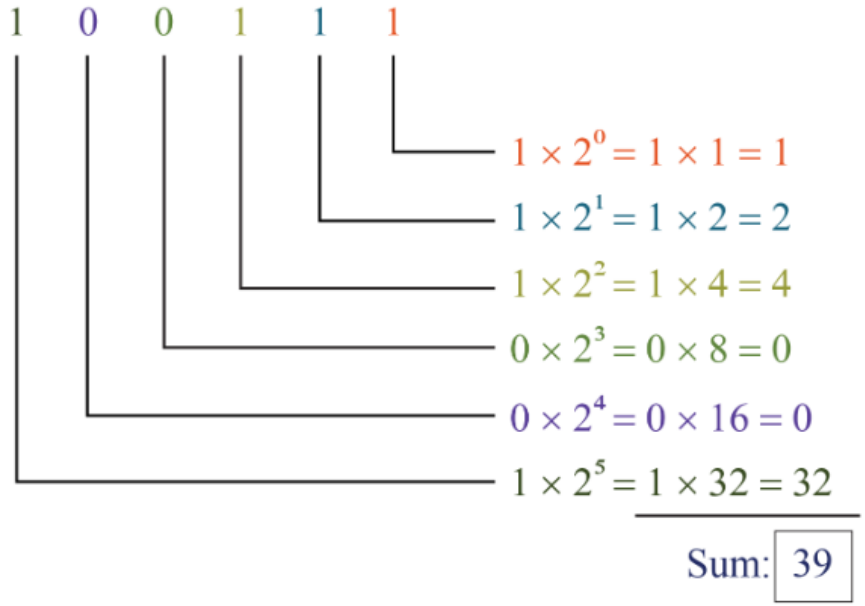
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Types of Number Systems

S.No.	Number System	Description	Example
1	Binary Number System	• Base 2. • Digits used: 0, 1	11010
2	Octal Number System	• Base 8. • Digits used: 0 to 7	125708
3	Hexadecimal Number System	• Base 16. • Digits used: 0 to 9 • Letters used: A- F	27FB
4	Decimal Number System	• Base 10. • Digits used: 0 to 9	1234

https://data-flair.training/blogs/number-systems-in-computer/

Binary → Decimal



$100111 = (1\times2^5) + (0\times2^4) + (0\times2^3) + (1\times2^2) + (1\times2^1) + (1\times2^0)$

$= (1\times32) + (0\times16) + (0\times8) + (1\times4) + (1\times2) + (1\times1)$

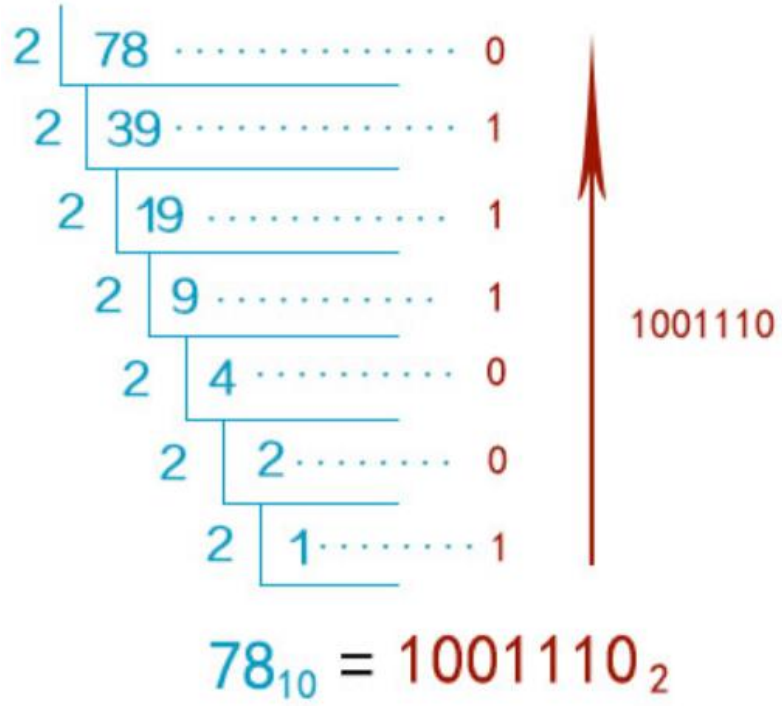
$= 32 + 0 + 0 + 4 + 2 + 1$

$= 39$

Thus, $100111_2 = 39_{10}$.

https://www.cuemath.com/numbers/number-systems/

Decimal → Binary



$78_{10} = 1001110_2$

https://www.goodconverters.com/decimal-to-binary

Hexadecimal → Decimal

1 b 7 e

$(1 \times 16^3) + (11 \times 16^2) + (7 \times 16^1) + (14 \times 16^0)$

4096 + 2816 + 112 + 14

7038

$1b7e_{16} = 7038_{10}$

https://www.inchcalculator.com/hex-to-decimal-converter/

Decimal → Hexadecimal

		Remainders
16 | 445		
16 | 27		13 → d
16 | 1		11 → b
0		1 → 1

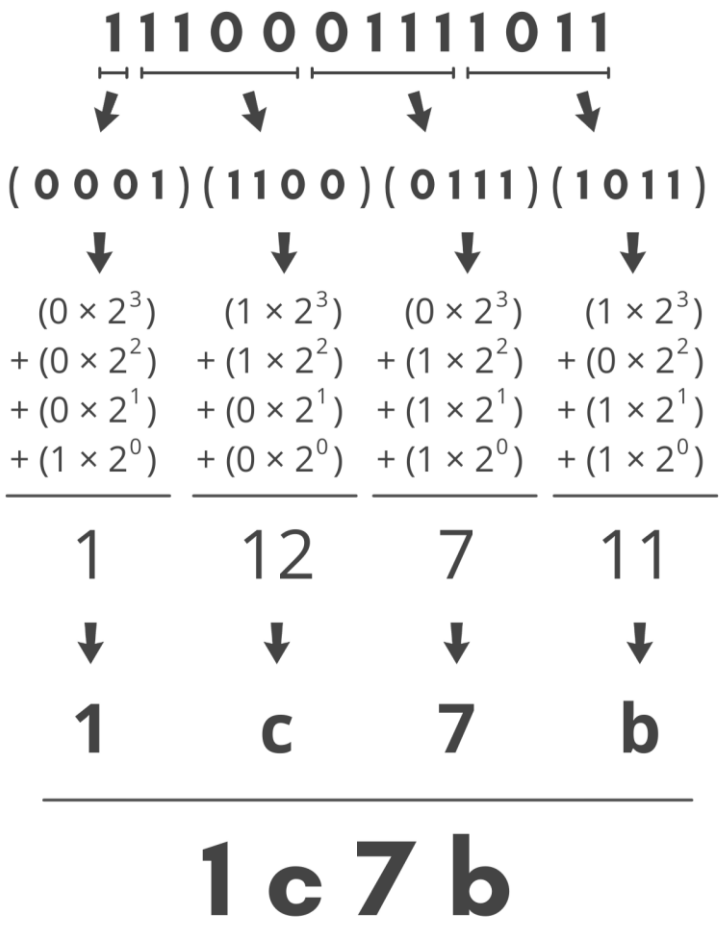
read remainders from bottom up

1bd

$445_{10} = 1bd_{16}$

https://www.inchcalculator.com/decimal-to-hex-converter/

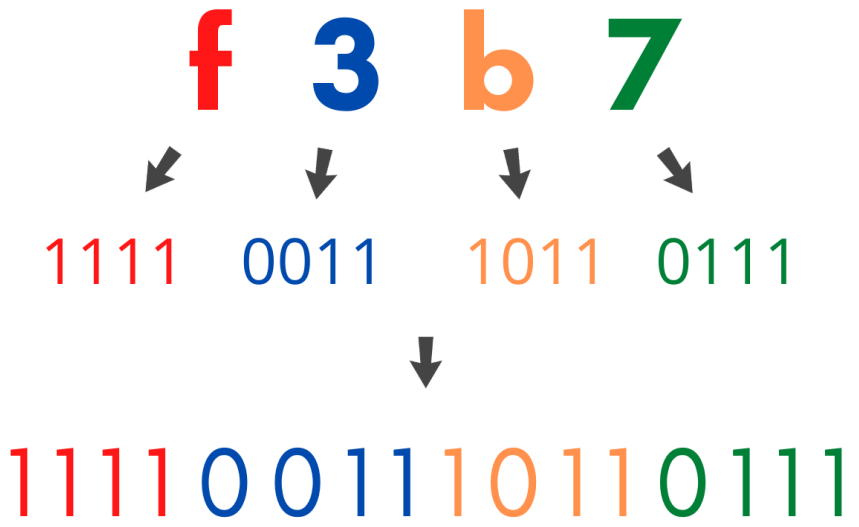
Binary → Hexadecimal



$1110001111011_2 = 1c7b_{16}$

https://www.inchcalculator.com/binary-to-hex-converter/

Hexadecimal → Binary



$f3b7_{16} = 1111001110110111_2$

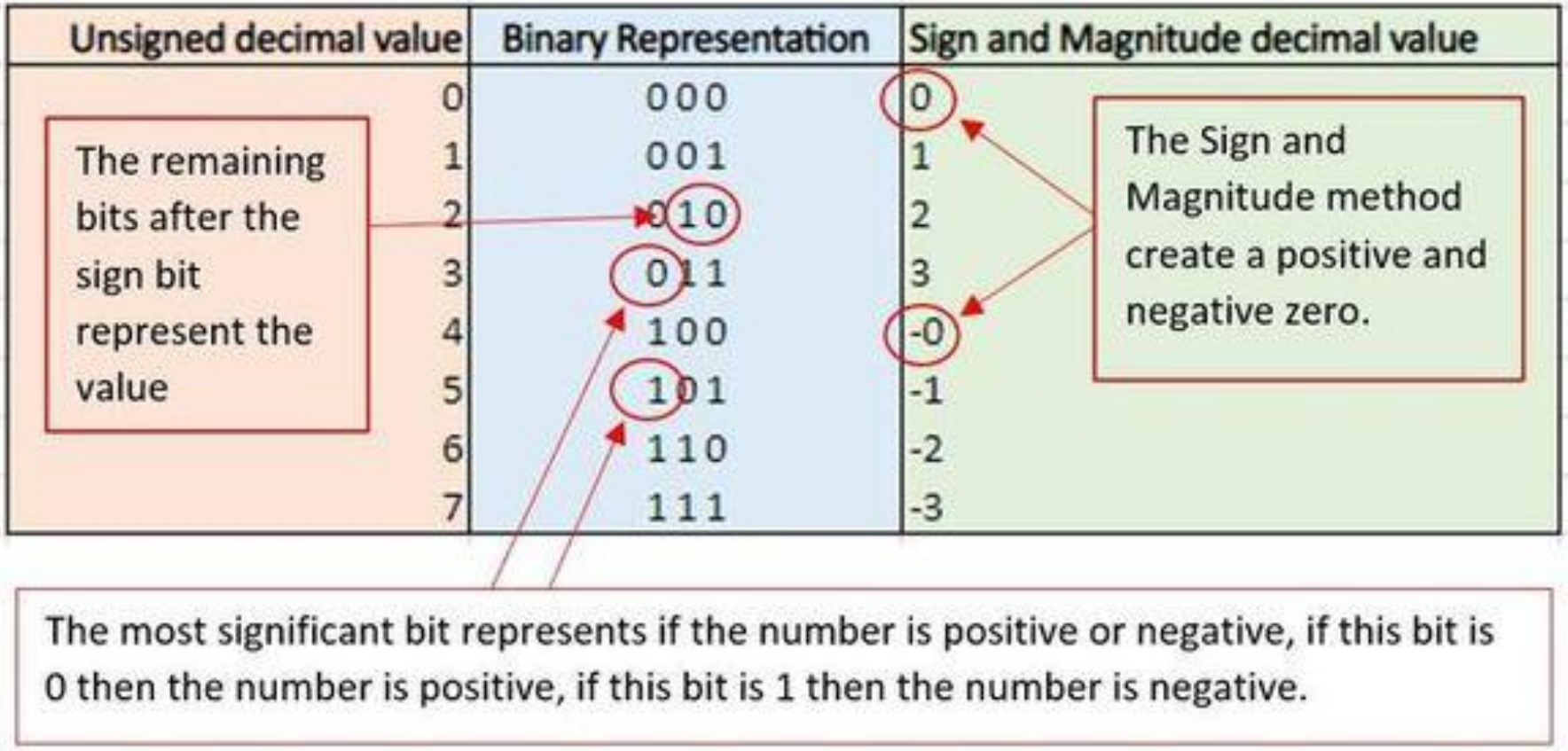
https://www.inchcalculator.com/hex-to-binary-converter/

Number System Relationship

HEXADECIMAL	DECIMAL	OCTAL	BINARY
0	0	0	0000
1	1	1	0001
2	2	2	0010
3	3	3	0011
4	4	4	0100
5	5	5	0101
6	6	6	0110
7	7	7	0111
8	8	10	1000
9	9	11	1001
A	10	12	1010
B	11	13	1011
C	12	14	1100
D	13	15	1101
E	14	16	1110
F	15	17	1111

https://www.tutorialspoint.com/basics_of_computers/basics_of_computers_number_system.htm

Negative Numbers: Sign and Magnitude



https://www.computersciencecafe.com/negative-number-representation.html

Negative Numbers: One's Complement

Unsigned decimal value	Binary Representation	One's Complement decimal value
0	000	0
1	001	1
2	010	2
3	011	3
4	100	-3
5	101	-2
6	110	-1
7	111	-0

Simply flip the bits to get the complement. 011 (+3) becomes 100 (-3) when the bits are flipped.

https://www.computersciencecafe.com/negative-number-representation.html

Negative Numbers: Two's Complement

Unsigned decimal value	Binary Representation	Two's Complement decimal value
0	000	0
1	001	1
2	010	2
3	011	3
4	100	-4
5	101	-3
6	110	-2
7	111	-1

Flip the bits then add 1 to get the complement

0 0 1 *Positive 1*
1 1 0 *Flip the Bits*
1 *Add 1 to the flipped value*
1 1 1 *Negative -1*

Note: Carry overs can be ignored

https://www.computersciencecafe.com/negative-number-representation.html